

Sai Kiran Kalyankar

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SUMMARY

Highly skilled and accomplished Senior Software Engineer with over 5.8 years of professional expertise in backend and full-stack development. Proficient in various technologies including Java/J2EE, Spring, Hibernate, and JavaScript/TypeScript (jQuery, Angular, React). Experienced in Microservice architecture using Spring MVC, Spring Boot, NestJS (Node.js), and ASP.NET, with configuration using Docker and Jenkins for continuous integration and build processes. Skilled in BDD, TDD and DDD using JUnit and Mockito, and experienced in cloud environments such as PCF, AWS, Azure and GCP. Extensive experience working on agile teams to deliver high-impact, scalable solutions in cloud platforms, while advocating for best practices in software engineering, usability, and maintainability. Proficient in front-end technologies like React and Angular and experienced in DevSecOps practices and AI/ML integration.

KEY HIGHLIGHTS

- 5+ years of Java/J2EE development expertise, including Java Collections Framework, RESTful APIs, JPA, and JUnit.
- 5+ years of Spring Boot experience in building microservices and scalable backend solutions.
- 3+ years of cloud experience in Pivotal Cloud Foundry (PCF), contributing to cloud-native applications.
- 2+ years of cloud experience in AWS and Azure.
- 1+ years of cloud experience in Google Cloud Platform (GCP).
- 2+ years of relevant experience Data Engineering and warehousing using Hadoop ecosystem tools and architecture.
- 2+ years of experience in C++ Programming.
- 1+ years of experience in .Net Programming.
- 2+ years of Python, AI/ML experience, including GenAI, RAG, Google Vertex AI, Chatbot hosted on Huggingface models, integrating AI/ML models into cloud applications.
- Strong knowledge in large-scale distributed system design and engineering best practices applying SOLID principles, design patterns, Microservices, 12-factor design, OOD.
- Expertise in creating responsive front-end applications from wireframes using Angular and React.
- Experienced working on SOA, full stack web, mobile, server, and database using multiple programming languages.
- Knowledge of data governance, security best practices, and implementing RBAC, encryption, and data masking, as well as authentication and authorization protocols such as OAuth, OAuth2, OpenID Connect, and SAML in cloud environments.
- Proven ability to work in agile environments with two-week iterations, ensuring timely delivery and adherence to best practices.
- Experience in the full SDLC including Scrum/Agile/Kanban and mob/pair programming (Xtreme Programming) methodologies, including experience applying Test-Driven Development (TDD), BDD, and Static Code analysis to improve quality and reliability along with security best practices including OWASP principles.
- Worked on Salesforce integration and POC projects using Apex Programming, Lightning Web Components (LWC), Visualforce, Triggers, SOQL, Controllers, Workflows, Custom config, and APIs for the development of CRM solutions.
- Strong knowledge in Machine Learning and Gen AI technology stack
- Strong understanding of DevOps tools like Git, Docker, Jenkins, and Gradle, with 4+ years of experience supporting CI/CD pipelines, advocating DevSecOps.

SKILLS & TOOLS

Languages: Java, TypeScript, Python, C#, C/C++, PHP, JavaScript, jQuery, Kotlin, Groovy, GoLang, JSP, JSF, Ajax, HTML/HTML5, CSS, Sass, SQL, NoSQL, GraphQL, TEX

Frameworks: Angular, Spring Boot, Hibernate, NestJS, .Net 7, NServiceBus, Jest, JUnit, Mockito, xUnit, Node.js, FastAPI, Flask, Hadoop

Libraries: React, Redux, ShadCN, TypeORM, LINQ, handlebarsjs, pug, OpenGL, WebGL, ThreeJS, OpenAI

Databases: MySQL, MSSQL, OracleDB, Postgres, MongoDB, SQLite, Firestore

Tools: Apache Kafka/Pulsar, Beeline, Hive, Impala, Oozie, BigQuery, Bigtable, DataProc, DataFlow, CloudFunctions, Cloud Run, Firebase, Figma,

DevOps and API Tools: Git, SVN, Docker, Kubernetes, Bash, Shell, GitLab, Bitbucket, Jenkins, Sonar, DAST, Swagger, Postman, Splunk

Cloud Skills: Pivotal Cloud Foundry (PCF), Amazon Web Services (AWS), Google Cloud Platform (GCP), Azure, IaC (Terraform)

Operating Systems: Windows, Linux, Unix, Android

IDEs: IntelliJ, NetBeans, Eclipse, STS, VS Code, Visual Studio, Android Studio, PyCharm

Other Skills: DevSecOps, Mobile/Android/Windows App development, UNIX, Powershell, AutoCAD, MATLAB, Salesforce

AI/ML: 3D Computer Vision and Graphics, Machine Learning, pandas, NumPy, Matplotlib, Seaborn, scikit-learn, CUDA, YOLO, NLP, NLTK, PyTorch, OpenCV, Ollama, Weka, RStudio, Rshiny, Data Analytics

WORK EXPERIENCE

Research Developer Lead

Wayne State University, Detroit, MI | Jul 2024 - Present

- Working as Lead Developer building a Full Stack Web Application Solution for Lipidomics, Pathology Department at WSU.
- Leading the development of application from scratch and continuous delivery of Features increasing accessibility and user engagement, utilizing a11y, WCAG standards.
- Designed intuitive front-end components using React and JavaScript to improve user experience, making applications more user-friendly and accessible.
- Created React based UI, FastAPI Backend for Lab Researchers to Access and Update the Lipids Sample data and restrictions using role-based fields and pages access control.
- Created React UI Calculator for external users to estimate the sub cost and total cost for sample analysis.
- Led the automation of CI/CD processes, reducing deployment time by 30%, which significantly enhanced the development workflow and productivity, Utilized Internal Platform CITADEL (C&IT Application DELivery) for Deployment.
- Developed a Retrieval-Augmented Generation (RAG) model-based AI chatbot using local LLM with Ollama and vector databases, enabling lab assistants to efficiently retrieve research data and answer queries in real-time without relying on external APIs.
- Designing and Developed website for research foundation conference
- Implemented and managed registration workflows, abstract submissions, and automated PDF eBook generation.
- Integrated Stripe payment processing to handle multiple registration categories and facilitate secure donations, optimizing transaction efficiency.
- Provided technical mentorship to junior developers and research staff, guiding them in best practices and professional growth

Stack: ReactJS, Python, FastAPI, SQLAlchemy, MySQL, GitLab, JS, Bootstrap, nginx, bash, Docker, CMS, Ollama, LLM, Stripe

Continuing Education Master's Degree USA

Aug 2022 - May 2024

Software Engineer Intern

Amrock - Rocket Companies, Detroit, MI | May 2023 - Aug 2023

- Full Stack Developer for Technocrats Platform Team to migrate monolith internal product (Atlas) – WCF, WPF to Cloud microservices using Azure and AWS, developing Workspace – Amrock Work Delivery System.
- Designed and developed a modern Angular frontend within an Nx Monorepo, utilizing:
 - Angular Material for a responsive, accessible, and consistent UI across the application.
 - Custom themes, dynamic forms, and data tables using Angular Material components such as MatTable, MatDialog, MatPaginator, and MatSort.
 - CDK Drag & Drop, Overlay, and Virtual Scrolling for an enhanced user experience.
 - State management using NgRx, optimizing performance and scalability.
 - Lazy loading and module federation for improved application performance.
- Implemented C#/ASP.NET backend services following the CQRS event-driven architecture and Domain-driven design (DDD), optimizing scalability and performance.
- Developed APIs along with regular code reviews, ensuring high quality and adherence to 100% compliance with security coding standards
- Created GraphQL services to enable efficient and flexible data querying and manipulation. These services allowed clients to request exactly the data they needed, reducing over-fetching and under-fetching of data, and improving overall application performance.
- Utilized .NET 7, Entity Framework, LINQ, and Stored Procedures for optimized data access, improving database performance and query execution efficiency.
- Integrated Azure DevOps CI/CD pipelines and containerized applications using Docker, ensuring smooth deployment workflows.
- Middleware components and custom exception handling for centralized error management and logging.
- Asynchronous programming (async/await, Task Parallel Library - TPL, Channels API) for improved scalability and responsiveness.
- Caching mechanisms (Redis, MemoryCache, Distributed Cache) to enhance API performance and reduce database load.
- Optimized database performance with Dapper and EF Core query tuning, reducing execution times and improving scalability.
- Secured APIs with CORS policies, CSRF protection, and security headers to enhance application security.
- Implemented API versioning and documentation with NSwag and Swagger UI for better maintainability and external consumption.
- Integrated Azure DevOps CI/CD pipelines and containerized applications using Docker, ensuring smooth deployment workflows.
- Implemented automated unit testing, integration testing, and API testing using:

- XUnit, NUnit, MSTest for unit and integration tests.
- FluentAssertions for more readable and maintainable assertions in unit tests.
- Moq and NSubstitute for mocking dependencies and improving test isolation.
- TestServer and WebApplicationFactory for in-memory API testing.
- Jest and Cypress for frontend and end-to-end testing.
- Leveraged Java 17's pattern matching for switch expressions to simplify complex business logic in one of Spring-boot based backend services, improving code readability and maintainability.
- Assisted in bug fixing, feature enhancements, and security vulnerability resolution, improving overall system stability.
- Configured logging and monitoring with Splunk to analyze system performance and detect anomalies in production environments.
- **Achievements:**
 - Received Amaze Recognition for creating team's GraphQL Service -Rocket Companies Hackweek Q2 -June 2023

Stack: Angular 15, C#, .Net 7, ASP.NET Core, Entity Framework, LINQ, Springboot, GCP, Azure DevOps, AWS, Docker, CQRS, SQL Server, Stored Procedures, GraphQL, Hot Chocolate, NserviceBus, Nx, Monorepo, NuGet, XUnit, Jest, Cypress, Snyk, Splunk

Continuing Education Master's Degree USA

Aug 2022 - May 2024

Analyst, Specialist, Application Development DBS Bank - Development Bank of Singapore - Hyderabad | Feb 2022 - Aug 2022

- Java Full Stack developer, for Consumer Banking for Investment and Insurance Project - FinTech using Frontend 20% and Backend 80%.
- Senior Backend Developer at SAILOR team (C2E) worked on Gamification Project (NavGames - To teach finances in an engaging way, by introducing a series of mini-games) using RESTful Microservices with Spring Boot.
- Contributed 20% to ReactJS development for the gamification of a financial quiz.
- Led 80% of backend development for gamification of a financial quiz with rewards, scores using Spring Boot for microservices, APIs, and low-level design (LLD) for SOAs and database management.
- Ownership of Product development for Internet Banking and Mobile Banking about financial planning and wealth management for Digibank.
- Collaborated with cross-functional teams, testing using GoMock, documented tech and design specifications.
- Enhanced security by implementing RBAC, OAuth, and encryption standards, protecting sensitive financial data.
- Presented project proposals and progress updates to higher management, conducted code reviews, and mentored junior developers, with Successful Go-Live, increasing user engagement by 30%.
- Led debugging efforts to improve application stability, resolving critical issues and enhancing system reliability.
- Optimized transaction workflows for improved response times, ensuring smooth and efficient banking operations.
- Enhanced system scalability by leveraging Java 11's improved CompletableFuture API for better asynchronous execution, optimizing transaction processing and reducing latency in banking operations.
- Implemented functional programming enhancements using Java 11's optional.orElseThrow() and new Collectors API methods, ensuring safer and more efficient handling of optional values in microservices.
- Optimized string manipulation using Java 17's enhanced text blocks, making SQL queries and JSON responses more readable and maintainable in investment and insurance applications.
- Leveraged Java 17's sealed classes to enforce strict inheritance rules, improving code maintainability and security for domain models in banking transactions.
- Integrated third-party APIs for financial data processing, expanding the system's capabilities and services.
- Conducted peer code reviews for quality assurance, ensuring high standards of code quality and performance.
- Wrote comprehensive unit and integration tests, ensuring robust and reliable software.
- Participated in knowledge-sharing initiatives within the development team, fostering a collaborative and learning-oriented environment.
- Ensured SDLC best practices were followed throughout development, maintaining high standards of software development.
- Microservices deployed and scaled on PCF Cloud Services.
- **Achievements:**
 - DBS TECH India Annual Award Top Engineering Team - SAILOR - C2E. -Aug 2022

Stack: Java 17, Spring Boot, ReactJS, Groovy, GCP, GoLang, Pivotal Cloud Foundry, Kubernetes, MariaDB, REST, GraphQL, Figma, Unit Test, Redis, Resilience4j, Zuul, Maven, Gradle, Bitbucket, Jira, Jenkins, SonarQube, postman

Technology Analyst

Infosys - Hyderabad | Apr 2021 - Feb 2022

- Client: Verizon

- JavaScript Full Stack developer as part of VZ Network Transparency ART (Service Assurance), Designed, developed, and maintained RESTful API endpoints.
- Led developing microservices including design and implementation of new products and features while enhancing existing product suite using JFrog Artifactory Repository Managers for builds and storing of Docker images.
- Created engaging UI components using JavaScript, React, and Node.js, enhancing the user interface and experience, Created charts and reusable components as per Verizon CSS standards, UI libraries for React, used Sass/SCSS, Handlebars for Components and Email Templates.
- Utilized TypeScript based Backend using NestJS Framework (Node.js Environment) to create microservices and implement Service-Oriented Architecture (SOA).
- Automated deployment pipelines using Jenkins and GitLab, streamlining the deployment process and reducing manual effort.
- Using an asynchronous distributed messaging and streaming platform (Apache Pulsar similar to Kafka) with security standards having Gov. and GIS data including data governance, security best practices, and implementing RBAC, encryption, and data masking using proxy microservice for sensitive data in cloud environments.
- Utilized Java 11's improved Concurrent API (CompletableFuture enhancements) to optimize asynchronous processing, improving the efficiency of distributed microservices handling Gov. and GIS data.
- Enhanced API performance and maintainability by leveraging Java 11 features, including var for local-variable type inference, reducing verbosity in RESTful API service implementations.
- Developed microservices with Customer Identity Access Management including Single Sign-On (SSO)/ OAuth
- Integrated PostgreSQL and NoSQL(MongoDB) databases for optimal performance, balancing the strengths of both types of databases, leveraging TypeORM and Mongoose ORM for schema modeling and data validation..
- Implemented data warehousing solutions using Hadoop, enabling efficient storage, processing, and analysis of large datasets.
- Using SAFe agile framework, collaborating with business and research teams in story grooming, reviewing story/acceptance criteria and Iteration planning.
- Utilized AWS EKS, RDS(Postgres), S3, Frontend 20%, Backend 80%, Achieved a 95% incident resolution rate through dedicated service ownership.
- Designed and executed test cases using JUnit and Jest, ensuring software reliability and performance.
- Implemented caching mechanisms for improved application speed, enhancing user experience and system performance.
- Debugged and resolved critical production issues, ensuring high system reliability and user satisfaction
- Deployed and maintained microservices on Google Cloud Platform with High availability to support the maintenance of services developed.
- Integrated Google Cloud Pub/Sub, Cloud Functions, and Cloud Run to streamline event-driven architecture
- Developed and managed containerized services on Google Kubernetes Engine (GKE).
- Provided mentorship to new hires and junior developers, guiding them in best practices and professional growth.
- **Achievements:**
 - Received Verizon Client Recognition and Appreciation for Echo to Kirke migration NT ART Project - Verizon Q1 - Feb 2022

Stack: ReactJS, NestJS, Spring boot, Java 11, GCP, NPM, NodeJS, ES6/7/8, TypeORM, SCSS, Pug, Pivotal Cloud Foundry, Docker, K8s, Helm, Jenkins, PostgreSQL, MongoDB, Hadoop, Jest, GIS, CI/CD, Apache Pulsar, Nexus, Sonar, Fortify, TwistLock, Blackduck, New Relic, Rally, Jira.

Engineer – Technology

Virtusa - Hyderabad | Dec 2017 - Apr 2021

- Client: BT (British Telecom), Lloyds Banking Group - Hyderabad

- Full Stack Java Developer, LAMP Stack Backend Developer using DevOps principles to add new functionality to LLUMS - Openreach's products.
- Developed application components in an Agile environment utilizing TDD to track E2E fiber build activity for Sitetracker (integrated with Salesforce) Squad, using Hadoop as a Service (HaaS) and Workflow Orchestration for Big Data Analytics.
- Developed ETL pipelines to ingest and transform data from various sources into the Hadoop ecosystem, ensuring data consistency and accuracy.
- Participated in major project migration, developing data models, pages, migration scripts, and optimizing systems to boost app performance by 50%.
- Performed database, SQL tuning and optimization for MySQL and Postgres achieving 99% uptime and reducing response time by 15% along with supporting Database Normalization.
- Upgraded and optimized applications using Java 11 features, including var for local-variable type inference, improving code readability and maintainability.

- Implemented Java 11's Functional Programming enhancements with optional.orElseThrow() and new Collectors API methods, simplifying null handling and improving stream processing efficiency.
- Optimized concurrent tasks with improvements using Java 11's CompletableFuture API, enabling efficient execution of asynchronous workflows within microservices.
- Migrated legacy code to leverage the new File API (Files.readString() and Files.writeString()), simplifying file handling operations and improving I/O efficiency.
- Deployed the spring microservice applications on PCF and scaled as needed using PCF services.
- Automated uploaders, inventory flows using SharePoint Excel File IO and SMTP notifications, enhancing monitoring and reducing manual effort for inventory systems by 30%
- Developed scripts to automate daily, weekly, and monthly cronjobs, including report generation and real-time batch-telemetry data feed processing using ETL and logging.
- Created Inventory Self Service Reports and billing reports using Spring JDBC and IBM MQ Messaging Queue including configuration, used JDK 1.8 features like Streams, Lambda, Optional to increase readability and performance.
- Generated reports in Excel, PDF files using the Apache POI and iText libraries.
- Involved in continuous enhancements using Frontend Angular 7, Spring JPA microservice, with Swagger/OpenApi for exposing API Metadata, including functional & non-functional requirements and fixing production issues.
- Design, implement, and manage access control mechanisms, including Policy-Based Access Control (PBAC), Role-Based Access Control (RBAC), Multi-Tenancy, and Attribute-Based Access Control (ABAC).
- Delivered results in complex cross-functional teams, presented, collaborated, and added value through documentation, debugging, test support and RCA.
- Participated in peer code reviews, mentored other engineers, providing comments and suggestions.
- **Achievements:**
 - Appreciation and tribe recognition for the work in "Agile Sitetracker squad" at BT -Nov 2020
 - Certificate of Appreciation for exemplary performance and contribution to "LLUMS team" at Virtusa. -Mar 2019
 - Client Appreciation with virtual client love badge. -Dec 2018

Stack: J2EE, Java 8, Java 11, Spring, Hibernate, ReactJS, Angular, SOAP and Restful web services, Ant, Maven, Junit, Mockito, MVC, Log4j, SLF4J, Netflix Zuul, Eureka, Hystrix, JS, Pivotal Cloud Foundry, jQuery, PHP, JSP, JSF, XML, MySQL, Oracle 10g, PL/SQL, MongoDB, Tomcat, Linux, Unix, Shell, bash, Git, GitLab, SVN, Jira, Citrix, Self-Service Views, Flyway, cron, IBM MQ, SOAP, Hadoop, hive, Avro, Cloudera HUE, oozie, ETL

EDUCATION

Wayne State University	Master's, Computer Science, AI Concentration Detroit, MI	Aug.'22 - May'24
Vardhaman College of Engineering	Bachelor's, Computer Science and Engineering Hyderabad, India	Sept.'13 - Jun.'17

RESPONSIBILITIES

- **Graduate Research Assistant** with Prof. Dr. Nic DePaula in tasks related to [LHEI](#) and his research around Gov. Health Administration by applying Data Analytics using Python, pandas, NLP for Twitter, Facebook Web Scraped Data and Full Stack Development using React for Heatmaps, Django backend – Wayne State University.
- **Lab Instructional Teaching Assistant** for Data Structures and Algorithms using C++, CSC 2200 Computer Science II at WSU, provided teaching, contributed to course assessment, mentoring, and grading support to 60 students.
- **College coordinator** and a member of CETA (Computer Engineer's Technical Association) – Vardhaman CS Department.
- **Organized** Web Designing Workshop event at TECHNOLITES'15 -Vardhaman's annual Techfest.

CERTIFICATIONS

- Salesforce Platform Developer 1
- Microsoft Technology Associate: Software Development Fundamentals.
- Certified SAgile 5 Agile Practitioner.
- Certified DBS Software Security for Developers.
- Certified DBS SRE Practitioner.
- CISSO: Certified Information Systems Security Officer.
- Completed ServiceNow Administrator Training

EXTRACURRICULAR ACTIVITIES:

- Participated in "**The Rise of AI Agents**" event by Google – Engaged in hands-on experience with AI agentic frameworks at the Google Detroit office, exploring advancements in AI automation and decision-making. – Aug 2024
- Attended the "**Data for Breakfast**" event by Snowflake – Explored Snowflake's cloud data platform, gaining hands-on experience with data pipelines, Cortex AI, and real-world enterprise data use cases – Mar 2025